

Hierarchical Bayesian Forecasting of U.S. State Unemployment Rates

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1. Introduction

Labour markets are central to economic well-being, and unemployment rates (UR) are key indicators of economic health. Recent disruptions like COVID-19, the 2008 Financial Crisis, and the 1980-1982 Volcker recession are nation-wide economic shocks but transmit unevenly across states depending on their individual economic strength, differing industries and more. For example, California is a rich tech and finance focused state and will respond differently to a recession than a state like North Dakota even under the same national shock. Therefore, accurate state-level unemployment forecasts are crucial for policymakers to design targeted interventions and allocate resources effectively.

Hierarchical Bayesian models are the natural solution here. Treating states in isolation ignores shared dynamics (no pooling) but averaging all states erases heterogeneity (full pooling). The hierarchical approach lets the data determine the degree of pooling through hyperparameters I additionally capture national shock transmission through macroeconomic covariates like the federal monetary policy (interest rates) and GDP growth. These variables serve as shared predictors across all states, allowing the model to separate common macro-driven movements from state-specific persistence.

The primary question this project answers is: Can a hierarchical Bayesian model that pools information across states produce more accurate unemployment forecasts than fitting each state independently?

Secondary question: Does including national macroeconomic covariates (federal funds rate, GDP growth) improve forecasts beyond an autoregressive model?

2. Literature Review

A paper by **Torabi (2012)** applies a hierarchical Bayesian model with MCMC to the same monthly BLS LAUS state-level unemployment data that I use. Their paper covers January 2004 – December 2007. Torabi’s model accounts for both spatial and temporal variation where state-level parameters are drawn from a hierarchical prior, spatial correlation between neighbouring states is captured via an intrinsic conditionally autoregressive prior, and temporal dynamics are modelled through a random walk. Their results are that hierarchical pooling substantially reduces estimation variance relative with the largest reductions for small states with unreliable sample sizes. This shrinkage behaviour is expected from partial pooling.

Our work differs from Torabi (2012) as Torabi’s goal is to improve estimation for a certain time period and is therefore using a small dataset whereas my goal is forecasting over a 24-month horizon using an AR(1) likelihood that explicitly models temporal persistence.

Additionally, Torabi captures cross-state dependence through geographic proximity while I capture it through shared national macroeconomic covariates since I want to focus on the notion that unemployment moves across states primarily through common macro shocks rather than geographic adjacency.

Torabi’s result that small states benefit most from pooling is what I expect as well. Posterior σ_β should be small,

indicating strong pooling, and states with more volatile or idiosyncratic unemployment histories (HI, NV, ND) should exhibit the most shrinkage toward the national mean.

3. Data

BLS LAUS (<https://www.bls.gov/lau/rdsncp16.htm>): monthly state-level unemployment rates for all 50 U.S. states. The time span is from January 1976 to December 2025 (311 months as October 2025 missing for all states).

Federal Reserve Economic Data (<https://fred.stlouisfed.org>):

I'll be using two national covariates that capture macroeconomic conditions:

- **Federal funds effective rate (FEDFUNDS) - Monthly:** This is the interest rate at which depository institutions trade federal funds with each other. It is a key tool of U.S. monetary policy that has a strong influence on economic activity and unemployment.
- **Real Gross Domestic Product (A191RL1Q225SBEA) - Quarterly:** This is the real gross domestic product, which is a measure of overall economic output. The growth rate of real GDP is closely related to unemployment.

Train/test split.

Training: January 1976 – December 2023 (28750 obs = 50 states × 48 years - 1 month (Oct 2025))

Test: January 2024 – December 2025 (1200 obs = 50 states × 24 months)

The training period spans three disruptions: national unemployment rate peaked at 10.8% (Volcker 1982), 10.0% (Great Financial Crisis 2008/2009), and 14.7% (COVID 2020).

4. Models

4.1 Model 1: Independent AR(1) — Naive Baseline

Each state is modelled independently with no pooling. For state s and time t :

$$\begin{aligned} \alpha_s &\sim \mathcal{N}(5, 3) && \text{(baseline level of unemployment is around 5\%)} \\ \beta_s &\sim \mathcal{N}(0.5, 0.3) && \text{(baseline autoregressive coefficient - moderate persistence)} \\ \sigma &\sim \text{Exp}(1) && \text{(shared SD across states)} \\ y_{s,t} \mid \alpha_s, \beta_s, \sigma &\sim \mathcal{N}(\alpha_s + \beta_s y_{s,t-1}, \sigma) \end{aligned}$$

Prior motivation: priors are *weakly informative* as they are centred on typical economic values (baseline UR $\approx 5\%$ and moderate AR persistence ≈ 0.5).

This model serves as the naive baseline. Since states are fit independently, estimates for small or volatile states will be noisy, with no regularisation from the national trend or neighbouring states. This model does allow states whose unemployment rates genuinely differ from the national picture to have their own parameters without being pulled toward the national mean.

4.2 Model 2: Hierarchical AR(1) with Covariates

State parameters are drawn from national hyperpriors, enabling partial pooling. In the hierarchical graphical model, α_s and β_s are themselves random variables whose distribution is determined by hyperparameters $(\mu_\alpha, \sigma_\alpha, \mu_\beta, \sigma_\beta)$. We use a non-centred parameterisation where instead of sampling $\alpha_s \sim \mathcal{N}(\mu_\alpha, \sigma_\alpha)$ directly, we

sample $\tilde{\alpha}_s \sim \mathcal{N}(0, 1)$ and set $\alpha_s = \mu_\alpha + \sigma_\alpha \tilde{\alpha}_s$. Macroeconomic covariates enter the likelihood as shared predictors:

$$\begin{aligned}
\mu_\alpha &\sim \mathcal{N}(5, 3) && \text{(national mean intercept)} \\
\sigma_\alpha &\sim \text{Exp}(1/2) && \text{(spread of intercepts across states)} \\
\mu_\beta &\sim \mathcal{N}(0.5, 0.3) && \text{(national mean persistence coefficient)} \\
\sigma_\beta &\sim \text{Exp}(1/0.3) && \text{(spread of persistence coefficients across states)} \\
\gamma &\sim \mathcal{N}(0, 1) && \text{(national economic covariate coefficients)} \\
\sigma &\sim \text{Exp}(1) && \text{(shared SD across states)} \\
\alpha_s \mid \mu_\alpha, \sigma_\alpha &\sim \mathcal{N}(\mu_\alpha, \sigma_\alpha) && \text{(state intercept, partially pooled)} \\
\beta_s \mid \mu_\beta, \sigma_\beta &\sim \mathcal{N}(\mu_\beta, \sigma_\beta) && \text{(state persistence coefficient, partially pooled)} \\
y_{s,t} \mid \alpha_s, \beta_s, \gamma, \sigma &\sim \mathcal{N}(\alpha_s + \beta_s y_{s,t-1} + \gamma^\top \mathbf{x}_t, \sigma)
\end{aligned}$$

where $\mathbf{x}_t = (\text{FEDFUNDS}_t, \text{GDP_growth}_t)^\top$ contains two national covariates, with `GDP\_growth` interpolated from quarterly to monthly.

5. Posterior Inference

5.1 Method 1: HMC via Stan (Primary)

Stan is used with HMC with 4 chains, 1000 warmup iterations, and 1000 sampling iterations per chain, yielding 4000 posterior draws. HMC is more efficient than Metropolis–Hastings as MH would degrade rapidly in this high-dimensional model.

5.2 Method 2: Variational Inference (Secondary)

Mean-field ADVI (`mod$variational, algorithm = "meanfield", draws = 1000`) is used on both models.

VI is used as a faster alternative to HMC as it converges in seconds compared to minutes for Stan’s sampler but this speed comes at the cost of the approximation quality, which I evaluate directly through comparison with HMC.

On Model 1, ADVI converges and the posterior means agree closely with HMC, but credible intervals are found to be tighter but this is expected because of the independence assumption $\prod_i q_i$ underestimating posterior variance.

On Model 2, mean-field ADVI fails entirely: the algorithm reaches the maximum number of dropped gradient evaluations before completing stochastic gradient ascent, and fullrank ADVI diverges after 10,000 iterations. The hierarchical posterior has strong dependencies between hyperparameters $(\mu_\beta, \sigma_\beta)$ and the state-level β_s ’s that no member of the mean-field family $\mathcal{Q}$ can represent, making the ELBO landscape ill-conditioned.

This is the key finding of the two-method comparison: HMC with non-centred parameterisation is necessary for Model 2, as VI is only viable for the simpler independent model.

6. Results

The following results are based on HMC inference, as VI fails on Model 2.

6.1 Convergence Diagnostics

Model 1: $\hat{R} \in [0.9995, 1.006]$, bulk ESS $\in [4489, 10991]$, zero divergences.

Model 2: $\hat{R} \in [0.9994, 1.009]$, bulk ESS $\in [578, 7378]$, zero divergences.

All $\hat{R} < 1.01$, indicating the 4 chains have converged to the same stationary distribution

6.2 Model 2 Hyperparameter Posteriors

6.3 Goodness of Fit and Shrinkage

We assess model fit using **posterior predictive checks**: for 50 draws $\theta^{(m)}$ from the posterior, we generate $y_{\text{rep}}^{(m)} \sim p(y|\theta^{(m)})$ and compare the predictive densities to the training data. Both models produce replicated data where marginal distributions closely match the observations, confirming neither model is misspecified.

The **shrinkage plot** compares $\hat{\beta}_s$ (posterior mean) from Model 2 versus Model 1 for all 50 states. All states fall below the diagonal so the hierarchical prior shrinks every state’s AR persistence toward $\mu_\beta = 0.966$. States with

Parameter	Mean	SD	5%	95%
μ_α (national mean intercept)	0.258	0.010	0.241	0.274
σ_α (spread of intercepts)	0.023	0.009	0.007	0.037
μ_β (national AR persistence)	0.966	0.002	0.963	0.968
σ_β (spread across states)	0.006	0.001	0.004	0.008
γ_{fed}	+0.006	0.001	0.005	0.007
γ_{gdp}	-0.034	0.001	-0.036	-0.033
σ	0.466	0.002	0.463	0.469

Table 1: Model 2 hyperparameter posteriors (HMC, 4000 draws).

$\mu_\alpha = 0.258$ is low relative to the prior mean of 5% UR, but the long-run mean implied by the AR(1) is $\mu_\alpha/(1 - \mu_\beta) = 0.258/0.034 \approx 7.6\%$, which is reasonable. $\sigma_\beta = 0.006$ indicates very strong pooling: state AR(1) persistence is nearly nationally determined. Both γ coefficients exclude zero at the 90% level and have the expected signs where higher fed rate predicts higher unemployment and higher GDP growth predicts lower unemployment.

the most extreme Model 1 estimates (HI, NH, NV) are shrunk most, consistent with the theory that noisier estimates receive stronger regularisation from the hyperprior.

6.4 Test-Set Forecasting

Model	Overall RMSE	Worst state
Model 1 (independent AR(1))	0.096	HI: 0.215
Model 2 (hierarchical + covariates)	0.118	WV: 0.164

Table 2: Posterior predictive mean RMSE on Jan 2024 – Dec 2025.

Model 2 achieves higher RMSE than Model 1. This reflects the bias-variance tradeoff inherent in partial pooling as $\sigma_\beta = 0.006$ aggressively shrinks all β_s toward the national mean, reducing estimation variance but introducing bias for states with different persistence patterns. This is not evidence of misspecification as both models pass PPC but illustrates that hierarchical shrinkage trades forecasting flexibility for estimation stability. This is consistent with Torabi (2012)’s observation that pooling gains are largest for small or noisy states.

6.5 Answering the Research Questions

Primary question: Partial pooling did not improve forecast accuracy on the 24-month test set (RMSE 0.118 vs. 0.096). The posterior $\sigma_\beta = 0.006$ shows that state AR(1) persistence is nearly nationally determined and is a substantial finding about labour market dynamics that the naive model cannot produce. The hierarchical model is a better estimator of underlying dynamics even when it is a worse forecaster in this period.

Additionally I assessed prior sensitivity by refitting Model 2 with $\sigma_\beta \sim \text{Exp}(1)$ (wider) and $\sigma_\beta \sim \text{Exp}(10)$ (tighter). All three priors yield a posterior mean of $\sigma_\beta \approx 0.006$, confirming the data dominates the prior and the strong pooling result is robust.

Secondary question: Both covariates are precisely estimated with economically meaningful signs ($\gamma_{\text{gdp}} = -0.034$, $\gamma_{\text{fed}} = 0.021$), confirming that national macroeconomic conditions contribute meaningful signal beyond pure AR(1) dynamics.

7. Discussion and Limitations

Prior sensitivity. Shocks like COVID can drive all states toward a common pattern as it’s a huge event and I can see that the result of $\sigma_\beta = 0.006$ is consistent with this. Analysis without 2020 is warranted.

VI failure and its implications. The ADVI failure on Model 2 is the most important methodological finding and highlights the importance of using a flexible inference method like HMC for complex hierarchical models.

References

Torabi, M. (2012). Hierarchical Bayes estimation of spatial statistics for rates. *J. Statistical Planning and Inference*, 142(2), 358–365.

Appendix: Code

A1. Model 1: Independent AR(1) (03_model1.ipynb)

```
library(cmdstanr); library(dplyr); library(ggplot2)
set.seed(405); TRAIN_END <- as.Date("2023-12-01")

bls <- read.csv("../cleaned_data/bls_data.csv") %>% mutate(date=as.Date(date))

bls_train <- bls %>% filter(date <= TRAIN_END) %>% arrange(state, date) %>%
  group_by(state) %>%
  mutate(bls_lag=lag(unemployment_rate), state_id=cur_group_id()) %>%
  filter(!is.na(bls_lag)) %>% ungroup()

states <- sort(unique(bls_train$state)); S <- length(states); N <- nrow(bls_train)

stan_data <- list(N=N, S=S, y=bls_train$unemployment_rate,
  y_lag=bls_train$bls_lag, state_id=bls_train$state_id)

mod1 <- cmdstan_model("model1_independent_ar1.stan")
fit1 <- mod1$sample(data=stan_data, seed=405, chains=4, parallel_chains=4,
  iter_warmup=1000, iter_sampling=1000, refresh=250)
fit1$save_object("../results/model1_fit.rds")

# Diagnostics
diag <- fit1$summary(c("alpha","beta","sigma")) %>%
  select(variable, mean, sd, q5, q95, rhat, ess_bulk, ess_tail)
print(paste("R-hat range:", paste(round(range(diag$rhat),4), collapse=" - ")))
print(paste("Bulk ESS range:", paste(round(range(diag$ess_bulk)), collapse=" - ")))

# Forecasting
alpha_hat <- fit1$summary("alpha")$mean; beta_hat <- fit1$summary("beta")$mean
bls_test <- bls %>% filter(date > TRAIN_END) %>% arrange(state, date)
test_dates <- sort(unique(bls_test$date))
y_prev <- bls_train %>% filter(date==TRAIN_END) %>% arrange(state) %>%
  pull(unemployment_rate)

forecast_list <- vector("list", length(test_dates))
for (i in seq_along(test_dates)) {
  y_pred <- alpha_hat + beta_hat * y_prev
  y_actual <- bls_test %>% filter(date==test_dates[i]) %>%
    arrange(state) %>% pull(unemployment_rate)
  forecast_list[[i]] <- data.frame(state=states, date=test_dates[i],
    y_pred=y_pred, y_actual=y_actual)
  y_prev <- y_actual
}
forecasts1 <- bind_rows(forecast_list)
print(round(sqrt(mean((forecasts1$y_pred - forecasts1$y_actual)^2)), 3))

# VI comparison
fit1_vi <- mod1$variational(data=stan_data, seed=405,
  algorithm="meanfield", draws=1000)
```

A2. Model 2: Hierarchical AR(1) (04_model2.ipynb)

```
fed <- read.csv("../cleaned_data/fedfunds_data.csv") %>% mutate(date=as.Date(date))
gdp <- read.csv("../cleaned_data/gdp_data.csv") %>% mutate(date=as.Date(date))

bls_cov <- bls %>% left_join(fed, by="date") %>% left_join(gdp, by="date")
bls_train <- bls_cov %>% filter(date <= TRAIN_END) %>% arrange(state, date) %>%
  group_by(state) %>%
  mutate(bls_lag=lag(unemployment_rate), state_id=cur_group_id()) %>%
  filter(!is.na(bls_lag)) %>% ungroup()

stan_data <- list(N=N, S=S, y=bls_train$unemployment_rate,
  y_lag=bls_train$bls_lag, state_id=bls_train$state_id,
  fed_rate=bls_train$fed_rate, gdp_growth=bls_train$gdp_growth)

mod2 <- cmdstan_model("model2_hierarchical_ar1.stan")
fit2 <- mod2$sample(data=stan_data, seed=405, chains=4, parallel_chains=4,
  iter_warmup=1000, iter_sampling=1000, refresh=250)
fit2$save_object("../results/model2_fit.rds")

# Forecasting
alpha_hat2 <- fit2$summary("alpha")$mean; beta_hat2 <- fit2$summary("beta")$mean
gamma_fed_hat <- fit2$summary("gamma_fed")$mean
gamma_gdp_hat <- fit2$summary("gamma_gdp")$mean

bls_test <- bls_cov %>% filter(date > TRAIN_END) %>% arrange(state, date)
y_prev <- bls_cov %>% filter(date==TRAIN_END) %>%
  arrange(state) %>% pull(unemployment_rate)

forecast_list2 <- vector("list", length(test_dates))
for (i in seq_along(test_dates)) {
```

```

covs <- bls_test %>% filter(date==test_dates[i]) %>% arrange(state)
y_pred <- alpha_hat2 + beta_hat2*y_prev +
  gamma_fed_hat*covs$fed_rate + gamma_gdp_hat*covs$gdp_growth
forecast_list2[[i]] <- data.frame(state=states, date=test_dates[i],
  y_pred=y_pred, y_actual=covs$unemployment_rate)
y_prev <- covs$unemployment_rate
}
forecasts2 <- bind_rows(forecast_list2)
print(round(sqrt(mean((forecasts2$y_pred - forecasts2$y_actual)^2)), 3))

# VI comparison
fit2_vi <- mod2$variational(data=stan_data, seed=405,
  algorithm="meanfield", draws=1000)

```

A3. Stan Model 1: Independent AR(1)

```

data {
  int<lower=1> N;
  int<lower=1> S;
  vector[N] y;
  vector[N] y_lag;
  array[N] int state_id;
}

parameters {
  vector[S] alpha;
  vector[S] beta;
  real<lower=0> sigma;
}

model {
  alpha ~ normal(5, 3);
  beta ~ normal(0.5, 0.3);
  sigma ~ exponential(1);

  for (n in 1:N) {
    y[n] ~ normal(alpha[state_id[n]] + beta[state_id[n]] * y_lag[n], sigma);
  }
}

generated quantities {
  vector[N] y_rep;
  for (n in 1:N) {
    real mu = alpha[state_id[n]] + beta[state_id[n]] * y_lag[n];
    y_rep[n] = normal_rng(mu, sigma);
  }
}

```

A4. Stan Model 2: Hierarchical AR(1)

```

data {
  int<lower=1> N;
  int<lower=1> S;
  vector[N] y;
  vector[N] y_lag;
  array[N] int state_id;
  vector[N] fed_rate;
  vector[N] gdp_growth;
}

parameters {
  real mu_alpha;
  real<lower=0> sigma_alpha;
  real mu_beta;
  real<lower=0> sigma_beta;
  real gamma_fed;
  real gamma_gdp;
  vector[S] alpha_raw;
  vector[S] beta_raw;
  real<lower=0> sigma;
}

transformed parameters {
  vector[S] alpha = mu_alpha + sigma_alpha * alpha_raw;
  vector[S] beta = mu_beta + sigma_beta * beta_raw;
}

model {
  mu_alpha ~ normal(5, 3);
  sigma_alpha ~ exponential(0.5);
  mu_beta ~ normal(0.5, 0.3);
  sigma_beta ~ exponential(0.3);
}

```

```

gamma_fed ~ normal(0, 1);
gamma_gdp ~ normal(0, 1);
sigma ~ exponential(1);
alpha_raw ~ normal(0, 1);
beta_raw ~ normal(0, 1);

for (n in 1:N) {
  real mu = alpha[state_id[n]] + beta[state_id[n]] * y_lag[n] + gamma_fed * fed_rate[n] + gamma_gdp * gdp_growth[n];
  y[n] ~ normal(mu, sigma);
}

generated quantities {
  vector[N] y_rep;
  for (n in 1:N) {
    real mu = alpha[state_id[n]] + beta[state_id[n]] * y_lag[n] + gamma_fed * fed_rate[n] + gamma_gdp * gdp_growth[n];
    y_rep[n] = normal_rng(mu, sigma);
  }
}

```